

E-PAPER: NOW JUST ROUND THE CORNER

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■ DR S.S. VERMA

Scientists and engineers had long predicted the advent of paperless offices, with widespread adoption of personal computers. Unfortunately, this never happened. Instead, personal computers caused consumption of more paper. It has now been realised that most of the paper consumption was caused by the differing comfort levels between reading documents on paper and reading them on the CRT screen.

Any document over a half-page length is likely to be printed, subsequently read and discarded. This leads to wastage of tonnes of paper. Thus, a need was felt for a paper-like electronic display—electronic paper (e-paper). It warranted as many paper properties as possible because ink on paper is the perfect display.

Thirty-five years in the making, e-paper is now closer than ever to changing the way we read, write and study—a revolution so profound that some see it as second only to the invention of the printing press in the 15th century. Made of flexible material, requiring ultra-low power consumption, cheap to manufacture, and most important, easy and convenient to read, e-papers of the future are just round the corner, with the promise to hold libraries in a chip and replace most printed newspapers

before the end of the next decade.

Paper-like features

E-paper is a display that possesses a paper-like high contrast appearance and is in a thin, light form. It gives the viewer the experience of reading from paper, while having the power to update information. Unlike a conventional flat panel display, which uses a backlight to illuminate its pixels, e-paper reflects light like ordinary paper and is capable of holding text and images indefinitely without drawing electricity, while allowing the image to be changed later. Due to qualities like stable images, its light

weight, durability, a wider viewing angle, and the fact that it uses reflected ambient light, e-paper is considered more comfortable to read than conventional displays. E-paper should not be confused with digital paper, which is a pad to create handwritten digital documents with a digital pen.

Combination of technologies

To build e-paper, several different technologies like electrophoretic and electro-wetting displays are used. Some technologies that use plastic substrate and electronics to make the display flexible are also used.

E-paper displays use technology enabled by electronic ink, which carries a charge, enabling it to be updated through



*An electronic paper display by Sony
(Photo courtesy: sonyinsider.com)*